

Vu A. Hoang

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EDUCATION

Troy University

Bachelor of Science in Computer Science

Troy, AL

Expected May. 2026

- **GPA: 3.59/4.00** | **Chancellor's List** | **International Trojan Opportunity Scholarship Recipient**

EXPERIENCE

The International Society of Data Scientists

Software Engineering Intern

Mar. 2025 – Present

Boston, MA (Remote)

- Optimized an AI-powered chatbot using **JavaScript/TypeScript** frontend and **FastAPI + LangChain** backend, improving UI responsiveness by 20%.
- Implemented advanced chatbot features including chat interruption handling, multi-format file ingestion, and multi-LLM support (OpenAI, Gemini, Anthropic) with user-provided API keys.
- Built automated Vietnamese legal document extraction and semantic search using **Selenium**, **BeautifulSoup**, **ChromaDB**, and **BKAI Vietnamese Bi Encoder**, saving **100+ hours** of manual work.
- Contributed to research on the paper “**Efficient Multi-Agent Collaboration for Medical Question Answering with Quantized LLMs under Resource Constraints**”, cited in ICITE 2025.

NAUCode

Software Engineering Intern

May. 2025 – Oct. 2025

Hanoi, Vietnam

- Developed reusable, component-driven UIs for two productivity-focused products using **React**, **TypeScript**, **TailwindCSS**, and **Ant Design**, reducing UI development time by **30%**.
- Built interactive frontend experiences for **HillChart** and **Solution Pitch**, including confidence-based task visualization, snapshot history, and guided problem-to-solution workflows.
- Contributed to backend development using **NestJS** and **Prisma**, integrating APIs and **Swagger documentation**, supporting authentication flows, ensuring smooth client-server data exchange.

Google Developer Group - Hanoi University of Science and Technology

Software Engineer

Sep. 2023 – Aug. 2024

Hanoi, Vietnam

- Co-developed the official homepage for a **100+ member** academic community using **Next.js**, **TypeScript** and **TailwindCSS**, resulting in a **20% increase** in online engagement and visibility.

PROJECTS

STD Information & Risk Assistant - [GitHub](#) | *React.js, FastAPI, ChromaDB, XGBoost, OCR*

Nov. 2025

- Led a team of 4 to develop a full-stack AI-powered sexual health assistant for Troy & Pike counties, using 25,000 synthetic patient records to provide personalized STD guidance with 70–75% ML prediction accuracy.
- Implemented a dual AI pipeline using **Retrieval-Augmented Generation (RAG)** retrieval and **XGBoost** classifier, achieving 85% citation compliance and 3–6 s average response time.
- Built with **React + TypeScript** frontend, **FastAPI** backend, **Supabase** database, **ChromaDB** embeddings, real-time **WebSocket** streaming, and persistent session history for scalable consultations.

Legal Document Crawler & Semantic Search Tool - [GitHub](#) | *Python, LangChain, ChromaDB*

Feb. 2025

- Automated the extraction and organization of Vietnamese legal documents (content, history, related articles, property, PDFs) using **BeautifulSoup** and **Selenium**.
- Enhanced semantic search using **ChromaDB** and **BKAI Vietnamese Bi Encoder**, achieving **60–70% top-5 relevance** for Vietnamese legal documents.

GDGoC - HUST Website - [Website](#) | *Next.js, TypeScript, TailwindCSS, Node.js, Vercel*

Mar. 2024

- Built and deployed a responsive, interactive website with **Next.js** and **TailwindCSS**, integrating Google Maps and an automated email service using **Resend** and **@react-email** for contact form submissions.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS, Java, C++, TypeScript

Frameworks & Libraries: React.js, TailwindCSS, FastAPI, Node.js, NestJS, Django, LangChain, XGBoost, PyTorch

Developer Tools: Git, GitHub, Docker, AWS, SQL Server, Figma, Vercel, Render

Certifications: Full-Stack Engineer (Meta), Machine Learning (DeepLearning.AI), Data Analytics (Google)